SMART STUDENT ATTENDANCE MANAGEMENT SYSTEM

Problem Statement:

Manual attendance systems are inefficient and prone to errors. This project provides a digital Java-based system to record, manage, and analyze student attendance automatically.

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TARGET USERS:

>> Teachers who need to efficiently record and monitor attendance
>> Academic administrators managing participation records
>> Educational institutions seeking digital attendance tracking

Key Classes & Responsibilities

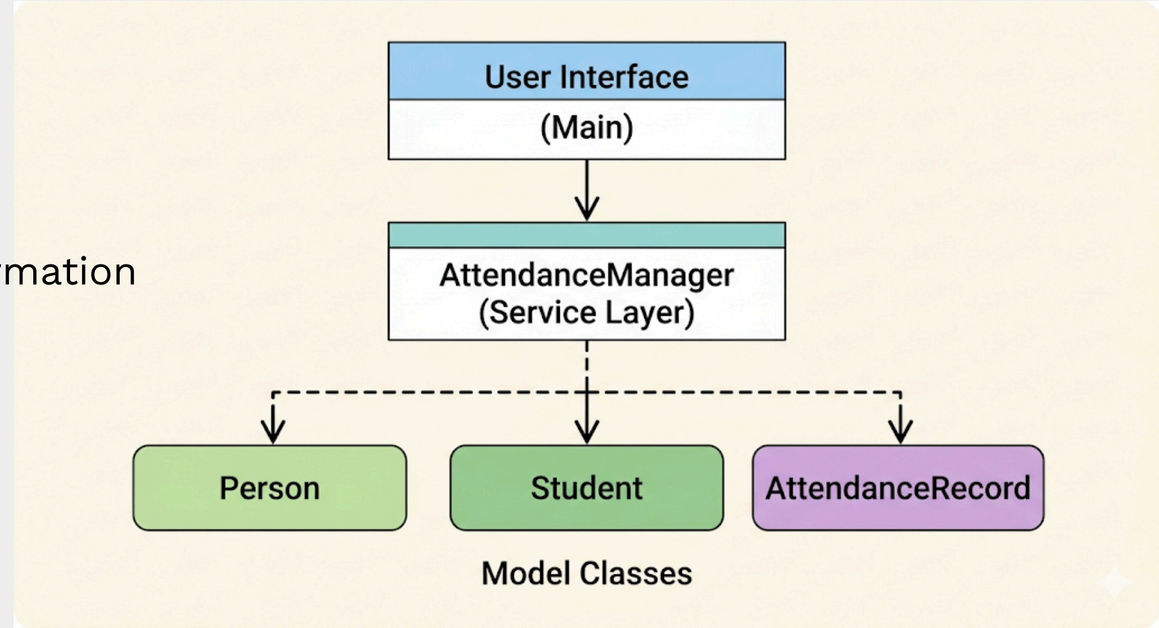
Person (Abstract Class)
- Contains common attributes: ID, Name

Student (Class)
- Inherits Person
- Stores student details and attendance information

AttendanceRecord (Class)
- Stores date and attendance status

AttendanceManager (Singleton Class)
- Registers students
- Marks attendance
- Calculates attendance percentage
- Manages records

Main Class
- Console interface with menu for users



OOP CONCEPTS USED

Abstraction

Abstract class Person defines common attributes like ID and Name.

Polymorphism

Methods such as displayInfo() or getDetails() are overridden in subclasses to provide specialized behavior.

Encapsulation

All class variables are declared private, accessed using getter and setter methods.

Inheritance

Student class extends Person, inheriting common attributes and behavior.

TECHNICAL HIGHLIGHT

Design Decision: Singleton Attendance Manager

The AttendanceManager class follows the Singleton pattern.
Why it was chosen:

- Ensures centralized management of attendance data
- Prevents creation of multiple attendance managers
- Improves data consistency
- Simplifies system architecture

1 Exception Handling

Custom exceptions are used:

- StudentNotFoundException
- DuplicateAttendanceRecordException

This ensures the system handles invalid operations gracefully without crashing.

2 Data Persistence

File handling is used to store:

- Student records
- Attendance data

This allows the system to load saved data when the application restarts.

SYSTEM WORKFLOW

Step 1 – Register Student

Enter Student ID: 101
Enter Name: Rahul
Student registered successfully

Step 2 – Mark Attendance

Enter Student ID: 101
Attendance: Present

Step 3 – View Attendance Percentage

Student: Rahul
Attendance: 85%

Error Handling

Invalid Student ID

Error : StudentNotFoundException
Student ID does not exist

>> This shows the system handles errors safely and prevents incorrect data entry.

THANK YOU